

# Session 5: Introduction to Vision Language Models

# Capability Need

- Apart from text, we want our models to understand images, videos and audios.
- Real-world intelligence is not one-dimensional.
- Various use cases- Medical scanning, solving mechanical problems, frontend coding tasks, etc.

Describe the pointed region in the image.



## GPT-4V:

The pointed region in the image is a row of hanging lights on a wire. The lights are small and round, and they are evenly spaced along the wire. The wire is stretched between two poles and is located above a patio area with tables and umbrellas. The lights appear to be decorative and are likely used to provide ambient lighting in the evening.

Describe the pointed region in the image.

| Method           | Validation set |      |       |      |       |      |         |      | Test set |      |       |      |       |      |         |      |
|------------------|----------------|------|-------|------|-------|------|---------|------|----------|------|-------|------|-------|------|---------|------|
|                  | in.            |      | near. |      | out.  |      | overall |      | in.      |      | near. |      | out.  |      | overall |      |
|                  | C              | S    | C     | S    | C     | S    | C       | S    | C        | S    | C     | S    | C     | S    | C       | S    |
| OSCAR            | 85.4           | 11.9 | 84.0  | 11.7 | 80.3  | 10.0 | 83.4    | 11.4 | 84.8     | 12.1 | 82.1  | 11.5 | 73.8  | 9.7  | 80.9    | 11.3 |
| Human            | 84.4           | 14.3 | 85.0  | 14.3 | 95.7  | 14.0 | 87.1    | 14.2 | 80.6     | 15.0 | 84.6  | 14.7 | 91.6  | 14.2 | 85.3    | 14.6 |
| VIVO             | 92.2           | 12.9 | 87.8  | 12.6 | 87.5  | 11.5 | 88.3    | 12.4 | 89.0     | 12.9 | 87.8  | 12.6 | 80.1  | 11.1 | 86.6    | 12.4 |
| ViViT            | 103.7          | 13.7 | 95.6  | 13.4 | 83.8  | 11.9 | 94.3    | 13.1 | 98.0     | 13.6 | 95.2  | 13.4 | 78.0  | 11.5 | 92.5    | 13.1 |
| UFO              | 103.9          | 14.5 | 95.5  | 13.8 | 83.5  | 12.3 | 94.3    | 13.6 | 98.9     | 14.3 | 94.7  | 13.9 | 77.9  | 12.1 | 92.3    | 13.6 |
| mPLUG            | -              | -    | -     | -    | -     | -    | 114.8   | 14.8 | -        | -    | -     | -    | -     | -    | -       | -    |
| SimVLM           | 113.7          | -    | 110.9 | -    | 115.2 | -    | 115.2   | -    | 113.7    | -    | 110.9 | -    | 115.2 | -    | 115.2   | -    |
| LEMON            | 118.0          | 15.4 | 116.3 | 15.1 | 120.2 | 14.5 | 117.3   | 15.0 | 112.8    | 15.2 | 115.5 | 15.3 | 110.1 | 13.7 | 114.3   | 14.9 |
| UniversalCap     | 123.2          | 15.0 | 121.5 | 15.3 | 123.4 | 14.4 | 122.1   | 15.0 | 118.9    | 15.4 | 120.6 | 15.3 | 114.3 | 14.1 | 119.3   | 15.1 |
| CoCa             | -              | -    | -     | -    | -     | -    | 122.4   | 15.5 | -        | -    | -     | -    | -     | -    | 120.6   | 15.5 |
| GIT <sub>B</sub> | 100.7          | 13.8 | 97.7  | 13.5 | 89.6  | 12.5 | 96.6    | 13.4 | -        | -    | -     | -    | -     | -    | -       | -    |
| GIT <sub>L</sub> | 107.7          | 14.9 | 107.8 | 14.5 | 102.5 | 13.7 | 106.9   | 14.4 | -        | -    | -     | -    | -     | -    | -       | -    |
| GIT              | 129.8          | 16.3 | 124.1 | 16.0 | 127.1 | 15.7 | 125.5   | 16.0 | 122.4    | 16.2 | 123.9 | 16.0 | 122.0 | 15.7 | 123.4   | 15.9 |
| GIT2             | 126.9          | 16.1 | 125.8 | 16.2 | 130.6 | 15.8 | 126.9   | 16.1 | 124.2    | 16.4 | 125.5 | 16.1 | 122.3 | 15.6 | 124.8   | 16.1 |

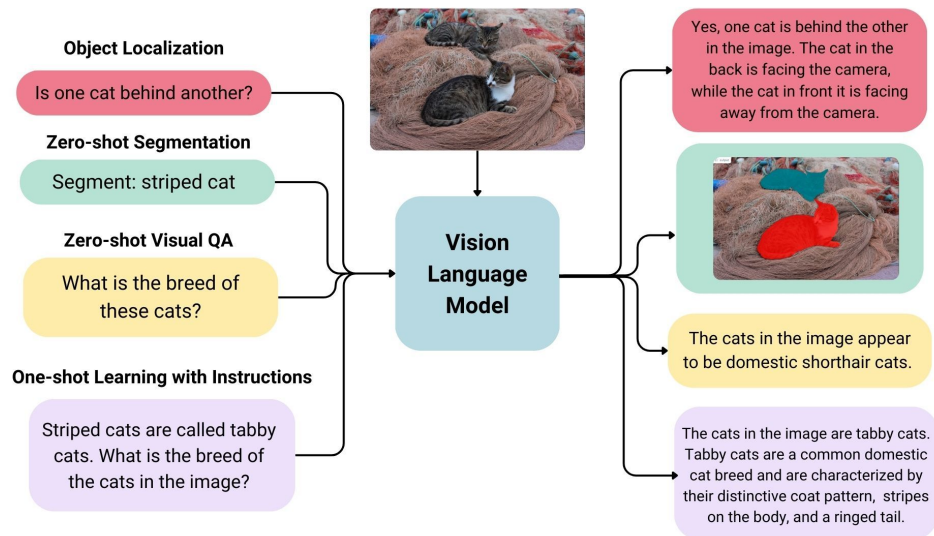
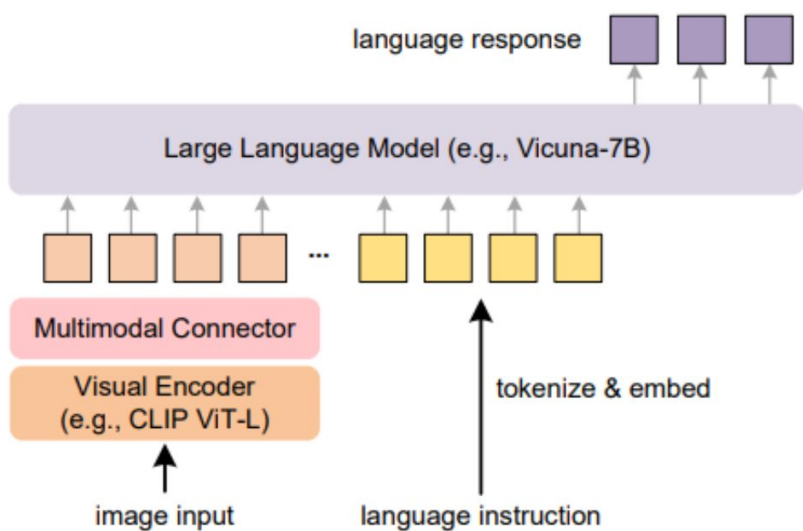
## GPT-4V:

The highlighted red region in the image is a table cell that contains a numerical value of 122.3. It is located in the "Test set" section of the table, under the "out" column and the "GIT2" row. This value represents the overall performance metric for the GIT2 method on the test set, in the "out" category.



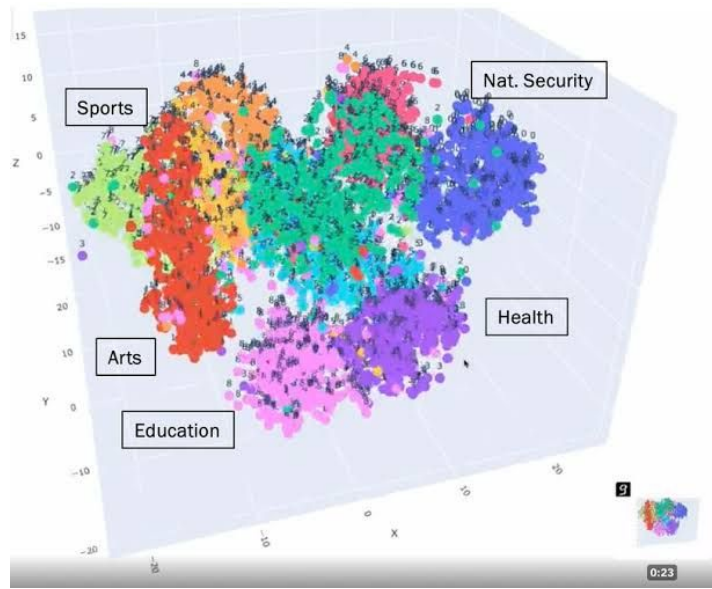
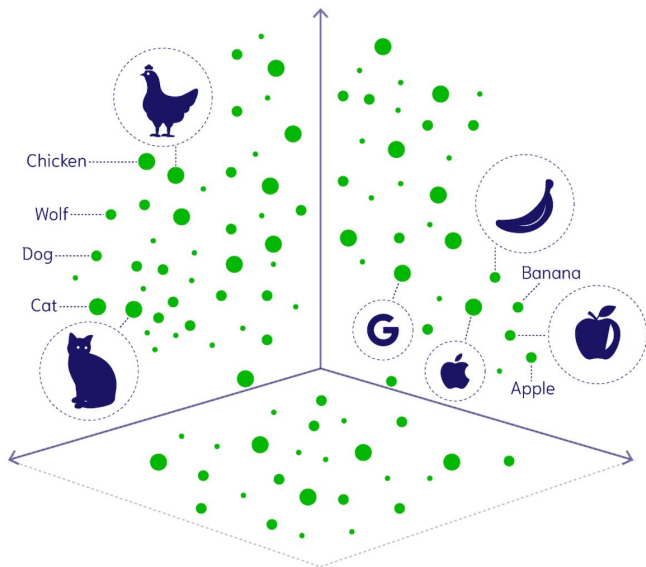
# Vision Language Model (VLM)

- Extends LLM capability by enabling it to understand and reason about images and text.
- VLM converts images into visual embeddings that are combined with text embeddings.
- Image captioning, OCR, charts, graphs, diagram understanding, etc.



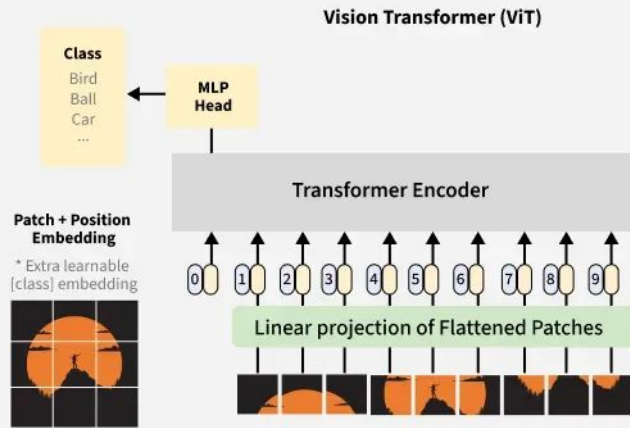
# Visual Embeddings

- Input Image -> Visual Encoder -> Visual Embeddings.
- A projector then maps the visual embeddings into the text embedding space (which is used by the LLM).
- Text and visual embedding representing same concepts lie close to each other.
- This allows LLM to treat image information just like text.

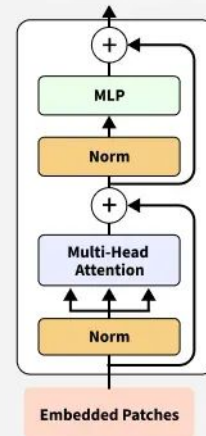


# Vision Transformer (ViT)

- Converts a raw image into a sequence of high-dimensional visual embeddings that the language model can process.



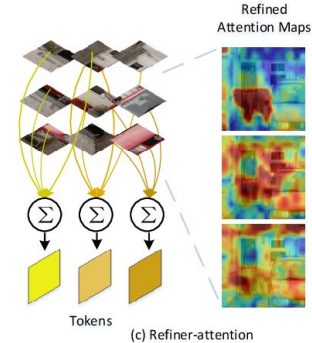
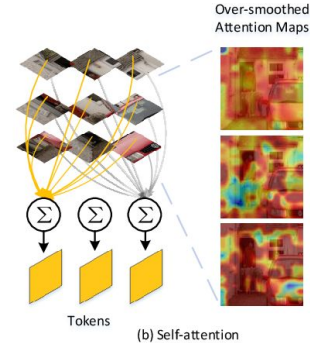
**Transformer Encoder**



**Image Patches**



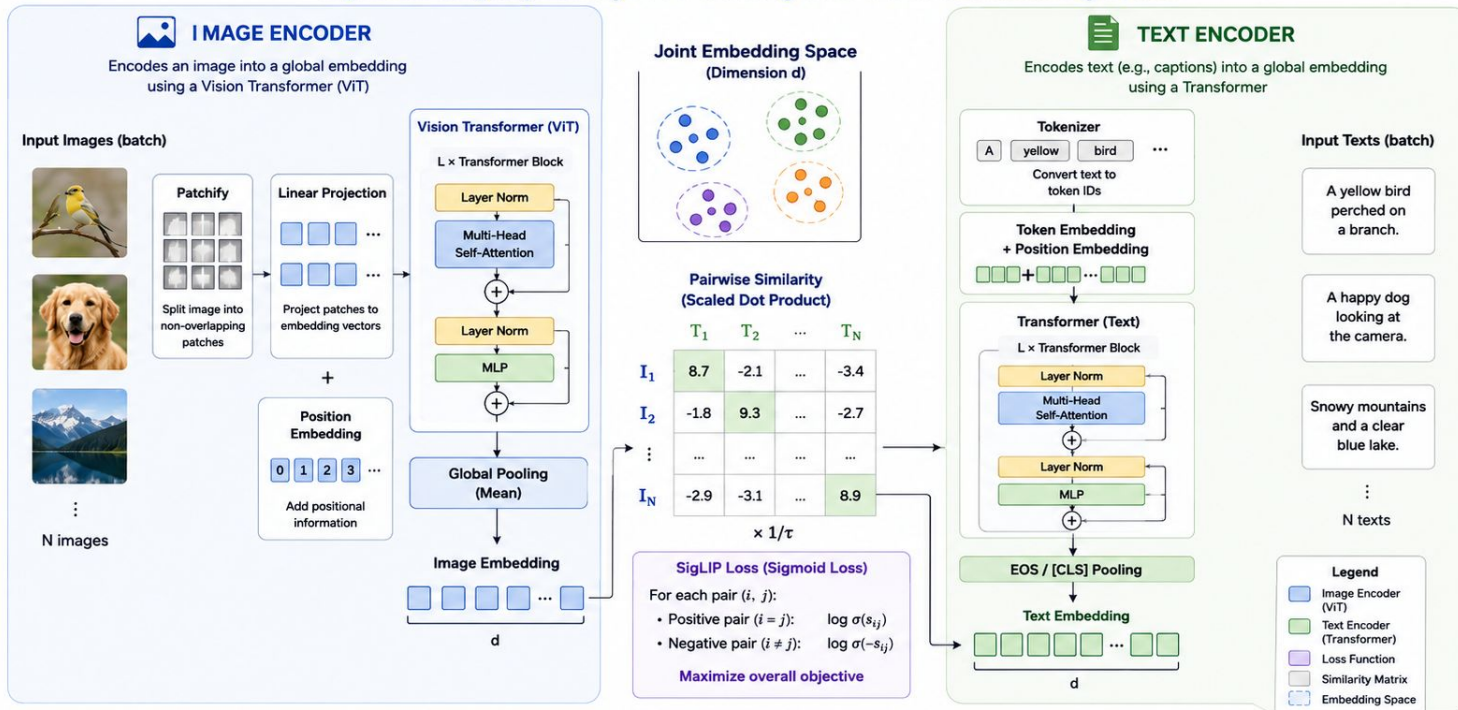
(a) Patch embedding





# SigLIP Architecture

## Sigmoid Language-Image Pre-training with a Joint Embedding Space



### How SigLIP Works



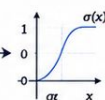
#### 1 Encode

Images and texts are encoded into dense embeddings of the same dimension.



#### 2 Compare

Compute pairwise similarity between all image and text embeddings.



#### 3 Sigmoid Objective

Use sigmoid loss to push matching pairs together and non-matching pairs apart.

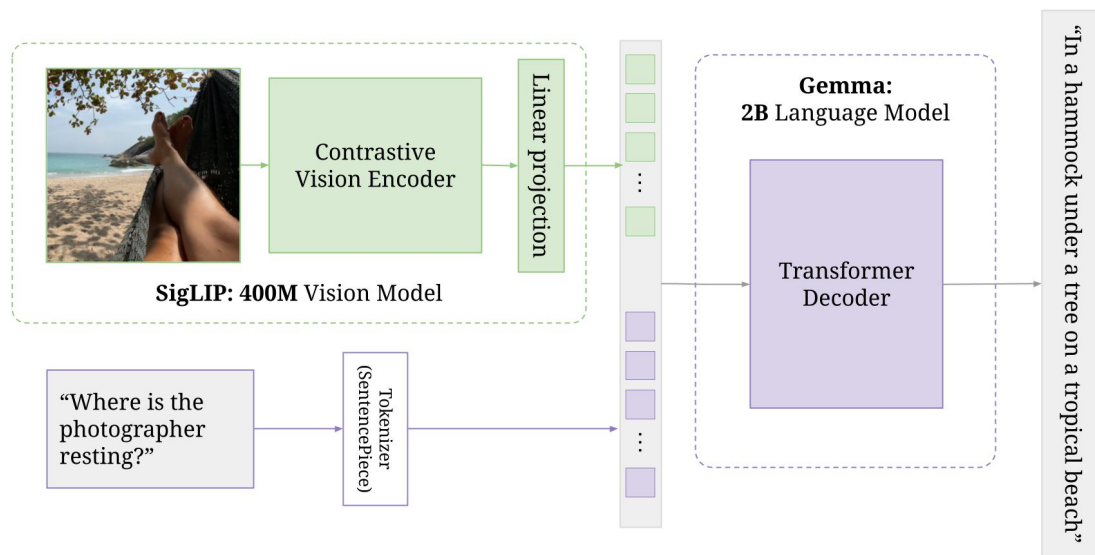


#### 4 Joint Space

Learn a shared embedding space where images and their captions are close.

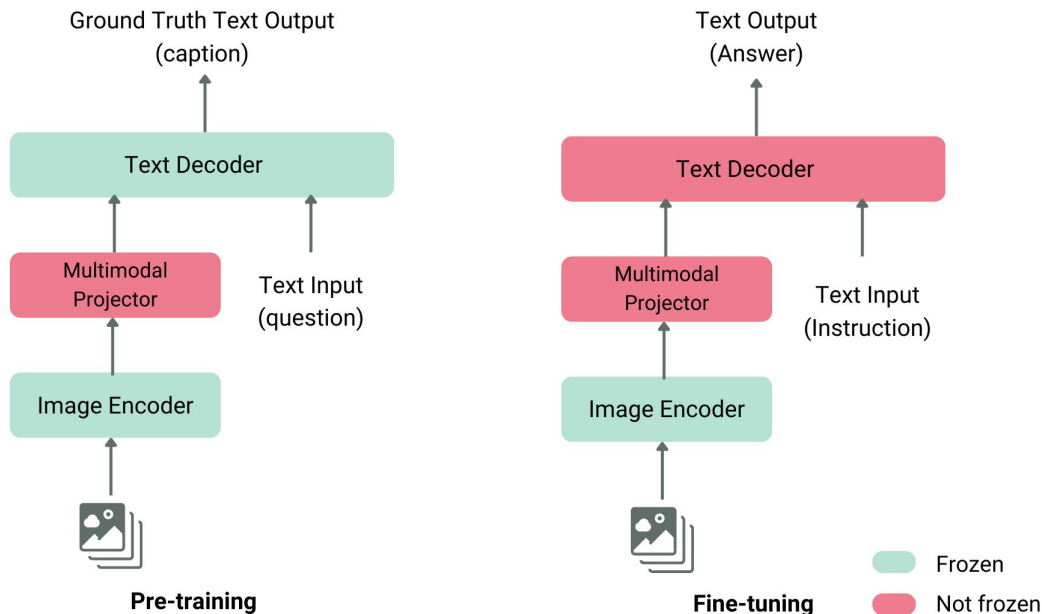
# Projector

- 2-layer MLP which connects vision encoder to the LLM. It has 2 jobs:
- 1. Match Dimension: Converts the vision encoder output dimension to the LLM model dimension  $\{d(\text{vision}) \rightarrow d(\text{LLM})\}$ .
- 2. Map visual features into the LLM text embedding space, so the LLM can interpret visual tokens alongside text tokens.



# Training a VLM

- Step 1: Pretraining -> Freeze vision encoder and LLM, train projector (2-Layer MLP).
- Step 2: SFT -> Freeze only vision encoder, train projector and LLM.





# Native Vision Models

- Models is designed and trained to process image in a unified architecture.
- Model learns vision-text relationship directly during training rather than a separately trained vision-language alignment layer.
- Gemini 3, ChatGPT-5, Claude-5, etc.

